test.py - C:\Users\rkash\OneDrive\Desktop\ram\test.py (3.14.7)

File Edit Format Run Options Window Help

```
import csv
import os
from datetime import datetime
import pandas as pd

FILE_NAME = "expenses.csv"

# Create CSV file if it doesn't exist
if not os.path.exists(FILE_NAME):
    with open(FILE_NAME, "w", newline="") as file:
        writer = csv.writer(file)
        writer.writerow(["Date", "Category", "Description", "Amount"])


# ---------------- VALIDATE DATE ----------------
def validate_date(date):
    try:
        datetime.strptime(date, "%Y-%m-%d")
        return True
    except:
        return False


# ---------------- ADD EXPENSE ----------------
def add_expense():
    date = input("Enter Date (YYYY-MM-DD): ")

    if not validate_date(date):
        print("Invalid date format!")
        return

    category = input("Enter Category: ")
    description = input("Enter Description: ")
```

```
    try:
        amount = float(input("Enter Amount: "))
    except:
        print("Invalid amount. Amount must be a positive number. Expense not saved.")
        return

    if amount <= 0:
        print("Invalid amount. Amount must be a positive number. Expense not saved.")
        return

    with open(FILE_NAME, "a", newline="") as file:
        writer = csv.writer(file)
        writer.writerow([date, category, description, amount])

    df = pd.read_csv(FILE_NAME)
    print(f"\nExpense saved. Total records: {len(df)}")


# ---------------- VIEW EXPENSES ----------------
def view_expenses():
    df = pd.read_csv(FILE_NAME)

    if df.empty:
        print("\nNo expenses found.")
    else:
        print("\n========== ALL EXPENSES ==========")
        print(df.to_string(index=False))


# ---------------- DELETE ALL EXPENSES ----------------
def delete_all_expenses():
    df = pd.read_csv(FILE_NAME)

    if df.empty:
        print("\nNo expenses found.")
```

```
    if df.empty:
        print("\nNo expenses found.")
        return

    confirm = input(
        "Are you sure you want to delete ALL expenses? (yes/no): "
    ).strip().lower()

    if confirm == "yes":
        with open(FILE_NAME, "w", newline="") as file:
            writer = csv.writer(file)
            writer.writerow(["Date", "Category", "Description", "Amount"])

        print("\nAll expenses have been deleted successfully!")
    else:
        print("\nOperation cancelled.")


# ---------------- CATEGORY SUMMARY ----------------
def category_summary():
    df = pd.read_csv(FILE_NAME)

    if df.empty:
        print("\nNo expenses found.")
        return

    summary = df.groupby("Category")["Amount"].sum()

    print("\n========== CATEGORY SUMMARY ==========")

    for category, amount in summary.items():
        if amount == int(amount):
            amount = int(amount)
        print(f"{category}: {amount}")
```

```
else:
    print("\nInvalid choice! Please try again.")
```

```
# ---------------- MAIN MENU ----------------
while True:

    print("\n=========================================")
    print("      PERSONAL EXPENSE TRACKER")
    print("=========================================")
    print("1. Add Expense")
    print("2. View Expenses")
    print("3. Category Summary")
    print("4. Monthly Summary")
    print("5. Delete All Expenses")
    print("6. Exit")
    print("=========================================")

    choice = input("Enter your choice: ")

    if choice == "1":
        add_expense()

    elif choice == "2":
        view_expenses()

    elif choice == "3":
        category_summary()

    elif choice == "4":
        monthly_summary()

    elif choice == "5":
        delete_all_expenses()

    elif choice == "6":
        print("\nThank you for using Personal Expense Tracker!")
        break
```

```
    total = df["Amount"].sum()

    if total == int(total):
        total = int(total)

    print(f"Total: {total}")


# ---------------- MONTHLY SUMMARY ----------------
def monthly_summary():
    df = pd.read_csv(FILE_NAME)

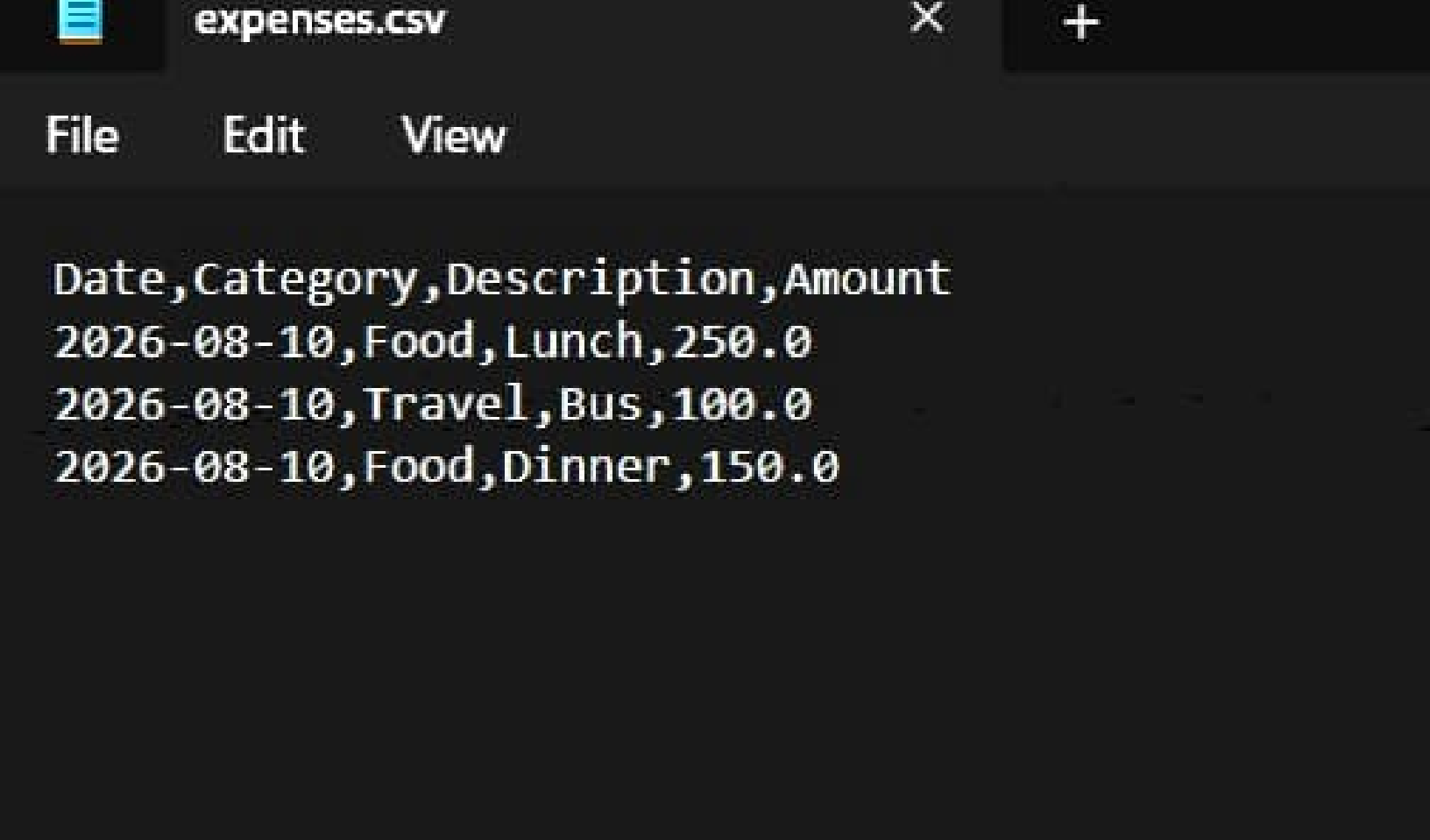
    if df.empty:
        print("\nNo expenses found.")
        return

    df["Month"] = pd.to_datetime(df["Date"]).dt.strftime("%Y-%m")

    summary = df.groupby("Month")["Amount"].sum()

    print("\n========== MONTHLY SUMMARY ==========")

    for month, amount in summary.items():
        if amount == int(amount):
            amount = int(amount)
        print(f"{month}: {amount}")
```


expenses.csv
File
Edit
View
Date,Category,Description,Amount
2026-08-10,Food,Lunch,250.0
2026-08-10,Travel,Bus,100.0
2026-08-10,Food,Dinner,150.0

```
      PERSONAL EXPENSE TRACKER
==============================================
1. Add Expense
2. View Expenses
3. Category Summary
4. Monthly Summary
5. Delete All Expenses
6. Exit
==============================================
Enter your choice: 1
Enter Date (YYYY-MM-DD): 2026-08-10
Enter Category: Travel
Enter Description: Bus
Enter Amount: 1

Expense saved. Total records: 2
```

```
      PERSONAL EXPENSE TRACKER
==========================================
1. Add Expense
2. View Expenses
3. Category Summary
4. Monthly Summary
5. Delete All Expenses
6. Exit
==========================================
Enter your choice: 1
Enter Date (YYYY-MM-DD): 2026-08-10
Enter Category: Food
Enter Description: Tea
Enter Amount: -50
Invalid amount. Amount must be a positive number. Expense not saved.
```

```
      PERSONAL EXPENSE TRACKER
==============================================
1. Add Expense
2. View Expenses
3. Category Summary
4. Monthly Summary
5. Delete All Expenses
6. Exit
==============================================
Enter your choice: 3

========== CATEGORY SUMMARY ==========
Food: 400
Travel: 100
Total: 500
```

```
==========================================
      PERSONAL EXPENSE TRACKER
==========================================
1. Add Expense
2. View Expenses
3. Category Summary
4. Monthly Summary
5. Delete All Expenses
6. Exit
==========================================
Enter your choice: 1
Enter Date (YYYY-MM-DD): 2026-08-10
Enter Category: Food
Enter Description: Lunch
Enter Amount: 250

Expense saved. Total records: 1
```